

ALEXANDER CORONEL

Software Engineer — Backend, Data & AI Integration

Quito, Ecuador | alexcoronel1995@gmail.com | +593 98 269 7669 | linkedin.com/in/acoronel95 | github.com/alexc957

SUMMARY

Software engineer with 6+ years of experience building and optimizing production systems for clients such as McKinsey & Company and Twilio. Specialized in PostgreSQL design and performance tuning, ETL pipelines, large-scale content and data migrations, and integrating LLMs into real products. Cut critical query times from ~10 s to under 1 s, led a Haskell-based survey platform, and work daily with agentic development workflows using Claude.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Haskell, Java

Backend & Frontend: Node.js, NestJS, Express.js, Django, Flask, React, React Native, Angular

Data & Cloud: PostgreSQL, MongoDB, MySQL, Microsoft Fabric, ETL, AWS (S3, RDS), systemd, Cron, Google Cloud Scheduler

AI & Tooling: OpenAI API, Hugging Face, Claude, GitHub Copilot, agentic development workflows

PROFESSIONAL EXPERIENCE

Software Developer — StackBuilders S.A.

Quito, Ecuador | Jul 2022 – Present

Survey Platform — Backend, Data & AI Systems (Haskell, PostgreSQL, Go)

- Led development of a large-scale Haskell survey platform for several months, owning architecture, technical decisions, implementation and stakeholder communication.
- Cut critical production query times from ~10 s to under 1 s by redesigning PostgreSQL schemas (new tables, views, targeted indexes) and tuning multi-table joins with EXPLAIN ANALYZE.
- Resolved production concurrency issues with PostgreSQL locking (FOR UPDATE), preventing competing processes from waiting needlessly on shared resources.
- Built a Go archival system, scheduled with systemd, that moved data older than 2 years (nearly 40% of stored data, unused by reports and application logic) to compressed archives in Amazon S3 and purged it from the operational database, improving database performance.
- Designed an OpenAI-based proof of concept for survey fraud detection that evaluates whether responses are contextually relevant to each question, with a Hugging Face model as fallback.

McKinsey & Company — Data Engineering & Analytics (Microsoft Fabric)

- Consolidated 9 KNIME workflows into 2 ETL pipelines in Microsoft Fabric (a ~78% reduction), simplifying the data-processing architecture.
- Designed a historical data model that replaced Excel-based data management and enabled scalable analytics and reporting in Tableau.
- Created AI agent skills that help data analysts understand the data model and build reports faster.
- Partnered directly with infrastructure owners and stakeholders to design and deliver the new processing and reporting architecture; contributed to a supplier diversity and inclusivity reporting platform.

Twilio — Documentation & Web Platform (Node.js, Django/Wagtail, AEM)

- Built a Node.js CLI that migrated ~600 pages (about 40% of a large documentation repository) from Wagtail and PostgreSQL to MDX, converting HTML, Markdown, code blocks and nested blocks while preserving page structure for a Docs-as-Code architecture.
- Developed features and reusable components on a large Django/Wagtail documentation platform, and built CLI tooling to move existing content to new page components.
- Improved multilingual content management by integrating and extending Wagtail Localize to fix gaps in the existing internationalization workflow.
- Maintained and extended Java components on Adobe Experience Manager (AEM) powering public product, pricing and blog content, resolving production issues.

AI-First Development Initiative (Claude)

- Took part in an internal agentic development initiative using Claude, working with contracts and Architectural Decision Records that define agent capabilities, limits, validation requirements and workflows.

- Contributed to automated ticket creation, code review and adversarial validation, where multiple agents independently evaluate each change.
- Conducted code reviews and mentored developers and interns across projects.

Quito, Ecuador | Aug 2021 – Jun
2022

Software Developer — Iqbit

- Built a full-stack debt tracking system, covering application development, backend services and data management.
- Developed and maintained a React Native radio app with real-time audio streaming.
- Automated recurring processes with Google Cloud Scheduler and Cron jobs, reducing manual processing by ~20%.

Quito, Ecuador | May 2019 – Nov
2019

Junior Web Developer — Mosflor SA

- Developed an e-commerce prototype with Angular and Flask for user testing and product feedback.
- Designed database schemas and technical documentation to improve developer onboarding.

EDUCATION

B.Eng. in Informatics and Computer Systems Engineering — Escuela Politécnica Nacional, Quito, Ecuador (Class of 2020)

LANGUAGES

Spanish: Native | English: Advanced (C1)